

Experimental Demonstration of Dual-Polarization PAM3 for Visible-Light Underwater Optical Wireless Communication

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Abstract We experimentally demonstrate dual-polarization PAM3 as a 3 bit/s/Hz IM/DD scheme for visible-light underwater optical wireless communication. Polarization-axis mismatch between transmitter and receiver deformed square-like constellation maps into rhombic ones, causing BER floor; affine compensation was effective for $\rho \leq 20^\circ$ but not at 30° . ©2026 The Authors

Introduction

Visible-light underwater optical wireless communication (VL-UOWC) is attractive for short-range high-speed links relevant to Beyond 5G/6G and underwater networking between remotely operated vehicles (ROVs), autonomous underwater vehicles (AUVs), and surface gateways. Because intensity modulation and direct detection (IM/DD) are practical with current visible-light devices, high spectral efficiency must be pursued without relying on coherent transceivers. However, simply increasing the modulation order of PAM rapidly tightens the required SNR [1]. Stokes-vector modulation, which exploits additional orthogonal dimensions in the received polarization state, has also been investigated [2,3]; however, it requires optical power splitting at the receiver, which lowers the received power per photodetector and is therefore disadvantageous in severely attenuation-limited UOWC.

In this work, we instead exploit two polarizations and experimentally demonstrate DP-PAM3, which provides 3 bit/symbol transmission using two directly modulated visible laser diodes. While this allows for higher data rates within the limited modulation bandwidth of current visible-light devices, the practical challenge is polarization-axis mismatch, which mixes the two polarization

channels and distorts the received constellation.

The main findings are as follows. First, DP-PAM3 was experimentally established in the present VL-UOWC testbed and achieved BER below the FEC limit. Second, polarization-axis mismatch deformed the received constellation into a rhombic shape and caused BER degradation with a clear error floor at large mismatch. Third, low-complexity static and dynamic affine compensation were effective up to about 20° , while a residual error floor remained at 30° .

Principle and Experimental Configuration

In the proposed DP-IM/DD UOWC system, two orthogonal polarization states are used to transmit two IM/DD signals simultaneously. If the two branches are modulated by OOK, PAM3, and PAM4, the corresponding dual-polarization formats provide 2, 3, and 4 bit/symbol transmission, respectively. In this paper, we focus on DP-PAM3.

Figure 1 illustrates the operating principle. Two Fabry-Perot laser diodes are aligned to orthogonal states of polarization and directly modulated by PAM3 waveforms. The joint constellation uses the eight two-dimensional intensity points obtained by omitting the (1,1) state, thereby realizing 3 bit/symbol. The state (1,1) is intentionally omitted to form an 8-point

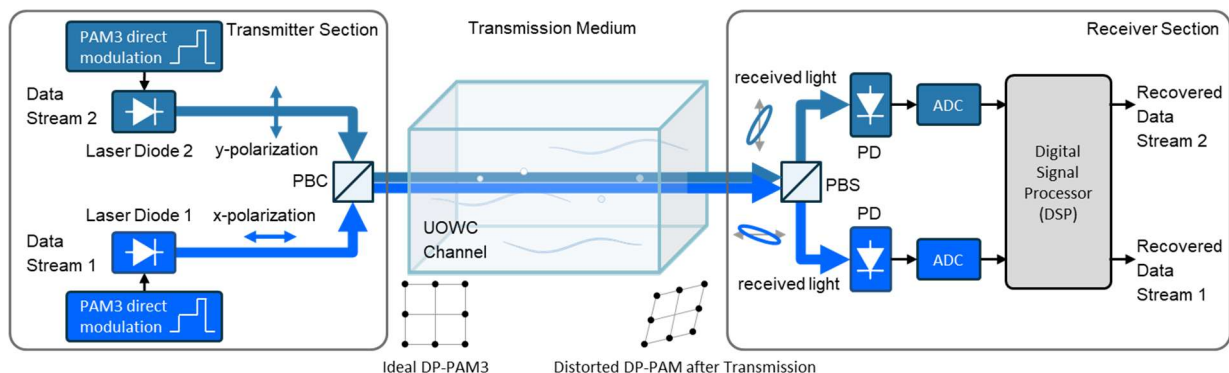


Fig. 1: Schematic of the proposed DP-PAM3 UOWC system.

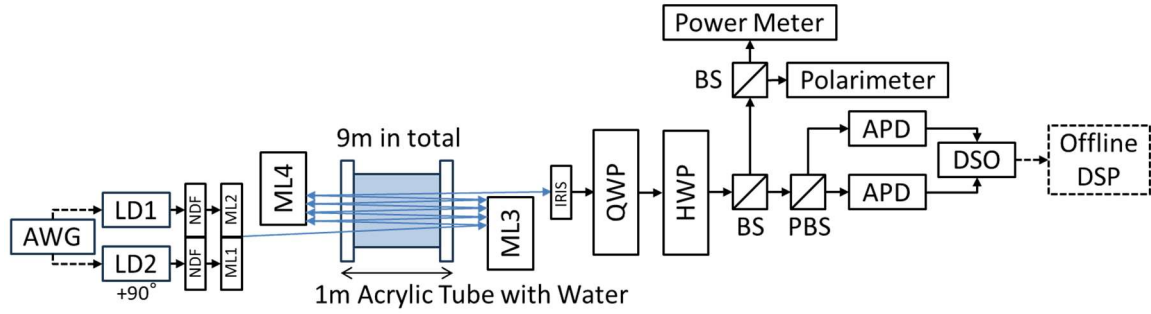


Fig. 2: Experimental setup used for BER evaluation of DP-PAM3 in the folded-path VL-UOWC link.

constellation, which effectively increases the average signal point distance. While the minimum Euclidean distance remains unchanged due to the remaining adjacent symbols, this geometry ensures that errors occur only between immediate horizontal or vertical neighbors. At the receiver, the two polarization branches are separated and jointly detected by nearest-neighbor detection in the two-dimensional IM/DD space.

Figure 2 shows the BER evaluation setup. Two FP-LDs (Thorlabs L450G2, around 450 nm) are driven by an arbitrary waveform generator (AWG) and launched with orthogonal polarization states. The folded-path propagation section includes a 1 m acrylic tube and mirrors, giving a total optical path of 9 m. At the receiver side, a quarter-wave plate (QWP) and a half-wave plate (HWP) are used before polarization separation, and the two electrical outputs are recorded by a DSO for off-line DSP.

Polarization characterization confirmed that the DOP change caused by filling the acrylic tube with water was less than 1% for both LD1 and LD2, whereas the ellipticity changed much more clearly. Therefore, the dominant polarization impairment in the present setup is not water-induced depolarization but a retardance-like polarization conversion introduced by the practical optical path. In the BER experiments, this setup-induced effect was compensated by inserting a QWP in the receiving optics before polarization separation.

Experimental Demonstration Results

The BER characteristics were evaluated for controlled polarization-axis mismatch values of $\rho = 0^\circ, 10^\circ, 20^\circ$, and 30° . For each received-power condition, three waveform records were processed off-line without a receiver matched filter, because the matched filter was found to degrade performance in the high-received-power region of the present experiment. Each record contained 2 million samples acquired with 10 samples/symbol oversampling, corresponding to 200,000 transmitted DP-PAM3 symbols. The BER values were averaged over three records, i.e., 600,000

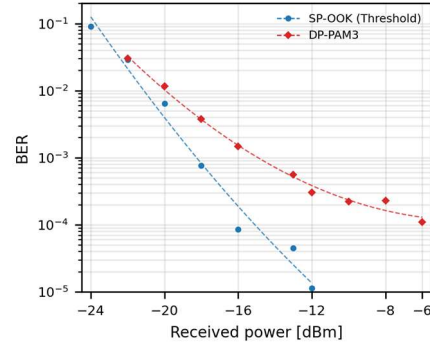


Fig. 3: BER comparison between SP-OOK detected by threshold and DP-PAM3 by nearest-neighbor at $\rho = 0^\circ$.

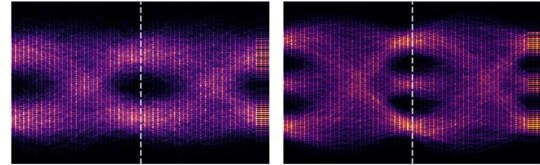


Fig. 4: Eye diagrams for SP-OOK (left) and DP-PAM3 (right) at BER around 10^{-3} .

transmitted symbols in total. We compare SP-OOK and DP-PAM3 and then evaluate three receiver conditions for DP-PAM3: no compensation, static or dynamic affine compensation.

Fig. 3 shows a BER comparison between SP-OOK and DP-PAM3 at $\rho = 0^\circ$. Both SP-OOK and DP-PAM3 achieved BERs below the FEC limit. Compared with SP-OOK, DP-PAM3 exhibited a penalty of approximately 3.5 dB at $\text{BER} = 10^{-3}$. This penalty is mainly attributed to the intrinsic modulation-format penalty caused by the increased number of signal levels, and it agrees well with the theoretical expectation. Fig. 4 shows eye diagrams for SP-OOK at -18 dBm and DP-PAM3 at -13 dBm, where both BERs are around 10^{-3} . Both schemes show clear eye openings.

Fig. 5 shows changes of constellation maps along with ρ . As ρ increases, the received DP-PAM3 constellation is linearly mixed and deformed from a square-like geometry into a rhombic one. This distortion directly causes BER degradation, and for large mismatch ρ a clear error floor appears. This observation is the key

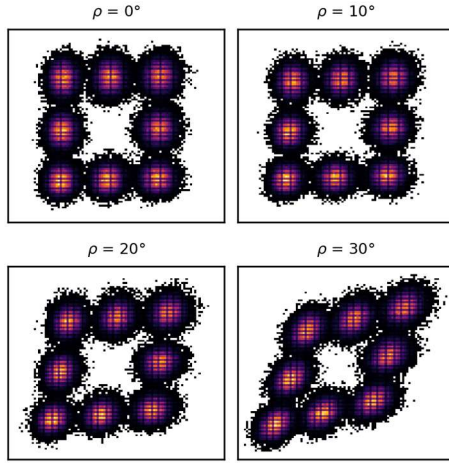


Fig. 5: Received DP-PAM3 constellation maps for different polarization-axis mismatch angles ρ .

experimental evidence that polarization-axis mismatch is the dominant impairment for practical DP-PAM3 UOWC links.

To mitigate this effect, we first apply static affine compensation. The received two-dimensional sample vector is modelled as $\mathbf{r} = \mathbf{A}\mathbf{s} + \mathbf{b} + \mathbf{n}$, where $\mathbf{A}$ is a 2×2 linear transform (distortion) matrix and $\mathbf{r}, \mathbf{s}, \mathbf{b}$ are received, transmitted, and offset sample vectors, respectively. By estimating $\hat{\mathbf{A}}$ and $\hat{\mathbf{b}}$ from received signal sample vectors, the effect of the polarization-axis mismatch can be compensated by estimating transmitted signal vector as $\hat{\mathbf{s}} = \hat{\mathbf{A}}^{-1}(\mathbf{r} - \hat{\mathbf{b}})$. This is a low-complexity linear correction converting rhombic constellation maps into square ones.

Fig. 6 shows BER characteristics for different polarization-axis mismatches without and with this simple affine compensation scheme. The results show that static affine compensation clearly improves BER for $\rho = 0^\circ, 10^\circ$, and 20° . At $\rho = 30^\circ$, the BER curve exhibits a clear error floor in the

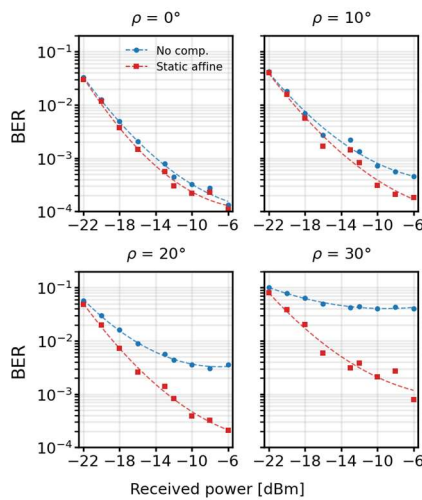


Fig. 6: BER performance of DP-PAM3 for different polarization-axis mismatch angles ρ with and without static affine compensation. Each point is averaged over 600,000 symbols.

high received-power region, while at 20° the residual degradation is milder and BER below 10^{-4} is still achieved. This indicates that the residual constellation distortion is no longer fully removed by the low-complexity affine compensation. Therefore, the 30° result should be interpreted not as a failure of DP-PAM3 itself, but as the boundary where a more advanced detection or polarization tracking becomes necessary.

We also evaluated dynamic affine compensation, in which $\hat{\mathbf{A}}$ and $\hat{\mathbf{b}}$ are estimated causally from M past symbols. Fig. 7 shows received power yielding $\text{BER}=10^{-3}$ against M . The horizontal solid and dashed lines are for the static affine compensation and no compensation, respectively. The results show that using about 50 to 100 past symbols provides the best performance and yields about 1 dB gain over the static affine compensation and nearly 2 dB gain over no compensation. When M becomes larger, the performance degrades slightly. At 30° , $\text{BER} = 10^{-3}$ was not reached within the measured received-power range even with the dynamic affine compensation, confirming that the large-mismatch regime requires a more advanced detection scheme.

Discussion and Conclusion

We experimentally demonstrated 1.8-Gbit/s DP-PAM3 transmission over a 600-MHz bandwidth, corresponding to 3 bit/s/Hz, in the present 9 m folded-path laboratory setup. While high-order SP-PAMs are constrained by the practical linear voltage swing of directly modulated LDs, DP-PAM3 achieves 3 bit/symbol with lower per-channel modulation levels by exploiting the polarization dimension. To cope with the polarization-axis mismatch, low-complexity affine compensation was effective for moderate mismatch up to about 20° , while a residual error floor remained at 30° . More advanced detection or polarization-tracking schemes will be investigated as future work.

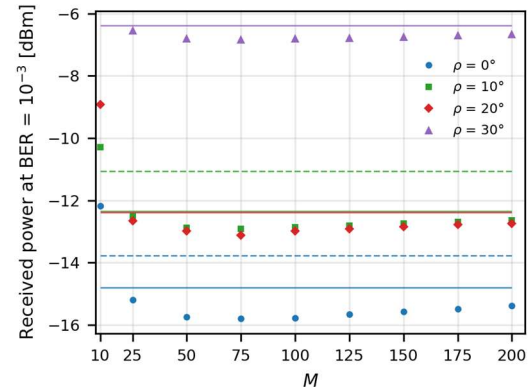


Fig. 7: Received power required to achieve $\text{BER} = 10^{-3}$ as a function of the number of past symbols M used for dynamic affine compensation. Solid and dashed horizontal lines indicate the static-affine and no-compensation results.

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References

- [1] E. Khansalee, Y. Okamura, and M. Hanawa, "Inverse nonlinear PAM4 modulation with high-power transmitter for extending transmission distances of underwater optical wireless communications," *Optics Communications*, vol. 577, 131421, 2025. DOI: [10.1016/j.optcom.2024.131421](https://doi.org/10.1016/j.optcom.2024.131421)
- [2] K. Kikuchi and S. Kawakami, "16-ary Stokes-vector Modulation Enabling DSP-based Direct Detection at 100 Gbit/s," *Optical Fiber Communication Conference*, paper Th3K.6, 2014. DOI: [10.1364/OFC.2014.Th3K.6](https://doi.org/10.1364/OFC.2014.Th3K.6)
- [3] K. Kikuchi, "Sensitivity analysis of optically preamplified Stokes-vector receivers using analytically derived formulae for bit-error rate," *Optics Express*, vol. 28, no. 18, pp. 26007-26017, 2020. DOI: [10.1364/OE.399814](https://doi.org/10.1364/OE.399814)